

- 8 (a) smaller deviation (not zero deviation) M1
 acceptable path wrt position of N A1
- (b) (i) mass of alpha particle = $4 \times 1.66 \times 10^{-27} \text{ kg}$ B1
 (kinetic energy = $0.5 \times 4 \times 1.66 \times 10^{-27} \times (1.30 \times 10^7)^2 \text{ J}$ A1
- (ii) all the kinetic energy becomes electrical potential energy B1
 $5.61 \times 10^{-13} = Q_{\alpha}Q_{\text{Au}} / 4\pi\epsilon_0 r = (2e) (79e) / 4\pi\epsilon_0 r$ C1
 $r = 6.48 \times 10^{-14} \text{ m}$ A1
- (c)
- emission from radioactive daughter products
 - self-absorption in source

- absorption in air before reaching detector
- detector not sensitive to all radiations
- window of detector may absorb some radiation
- background radiation

Any two points.

B2

- (d) (i) curve is not smooth or curve fluctuates/curve is jagged B1
(ii) clear evidence of allowance for background B1
half-life determined at least twice B1
half-life = 1.5 hours A2
(2 marks if in range 1.4 – 1.6; 1 mark if 1.6 < half-life ≤ 2.0)
- (e) 1. half-life: no change M1
because decay is spontaneous/independent of environment A1
2. count rate (likely to be or could be) different / is random / cannot be predicted B1
- (f) activity = $(3.8 \times 10^4) e^{(-\ln 2 / 15)(36)}$ C1
or activity = $(3.8 \times 10^4) [1 / 2^{2.4}]$ (C1)
= 7200 Bq A1
volume = $(7200 / 1.2) \times 5.0 = 3.0 \times 10^4 \text{ cm}^3$ A1
or activity of $5.0 \text{ cm}^3 = 1.2 \times 2^{2.4}$ (C1)
= 6.3336 Bq (C1)
volume = $(3.8 \times 10^4 / 6.3336) \times 5.0 = 3.0 \times 10^4 \text{ cm}^3$ (A1)